YeshwantraoChavan College of Engineering
(An Autonomous Institution affiliated to RashtrasantTukadojiMaharaj Nagpur University)
Accredited 'A++' Grade by NAAC
Hingna Road, Wanadongri, Nagpur-441110

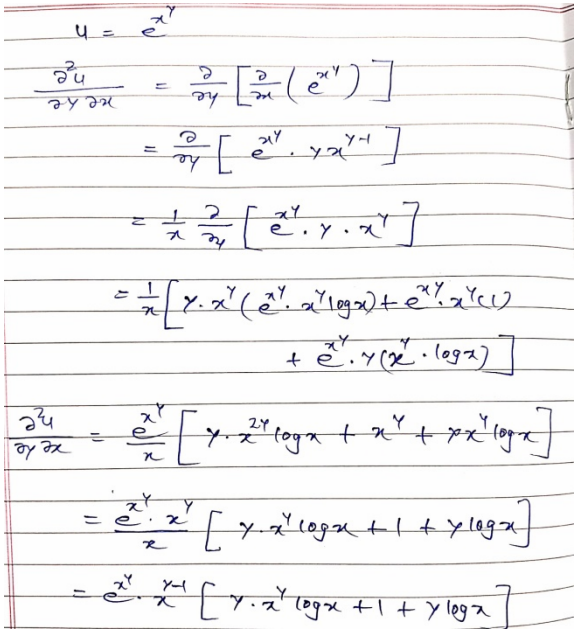
Department of Applied Mathematics & Humanities

Electronic Question Answer Bank

Course Name: Calculus and Vector (C & V)

Course Code: 23GE1101

UNIT-II: PARTIAL DIFFERENTIATION

Sr.No.	Unit No.	Text of Question	Time required in Minutes	Marks allotted	lengthiness	Difficulty level	CO Mapping	Type Descriptive / Numerical
1	Unit-II	$u = e^{x^y}$, find $\frac{\partial^2 u}{\partial y \partial x}$ Sol: 	15	5	Low	Easy	CO-2	Descriptive l
2		If $u = \tan^{-1}\left(\frac{x^3+y^3}{x-y}\right)$, find the value of	15	5	Low	Easy	CO-2	Descriptive

	Unit-II	Sol:						

$\Rightarrow x.\frac{\partial^2 u}{\partial x^2}+y\frac{\partial^2 u}{\partial x\,\partial y}=(2\cos 2u-1)\frac{\partial u}{\partial x}$

Multiplying with x, we get

$$x^2\frac{\partial^2 u}{\partial x^2}+xy\frac{\partial^2 u}{\partial x\,\partial y}=x(2\cos 2u-1)\frac{\partial u}{\partial x} \quad \ldots..(2)$$

Again differentiating (1) partially w. r. to y , we have

$$x.\frac{\partial^2 u}{\partial y\,\partial x}+y\frac{\partial^2 u}{\partial y^2}+\frac{\partial u}{\partial y}=2\cos 2u\frac{\partial u}{\partial y}$$

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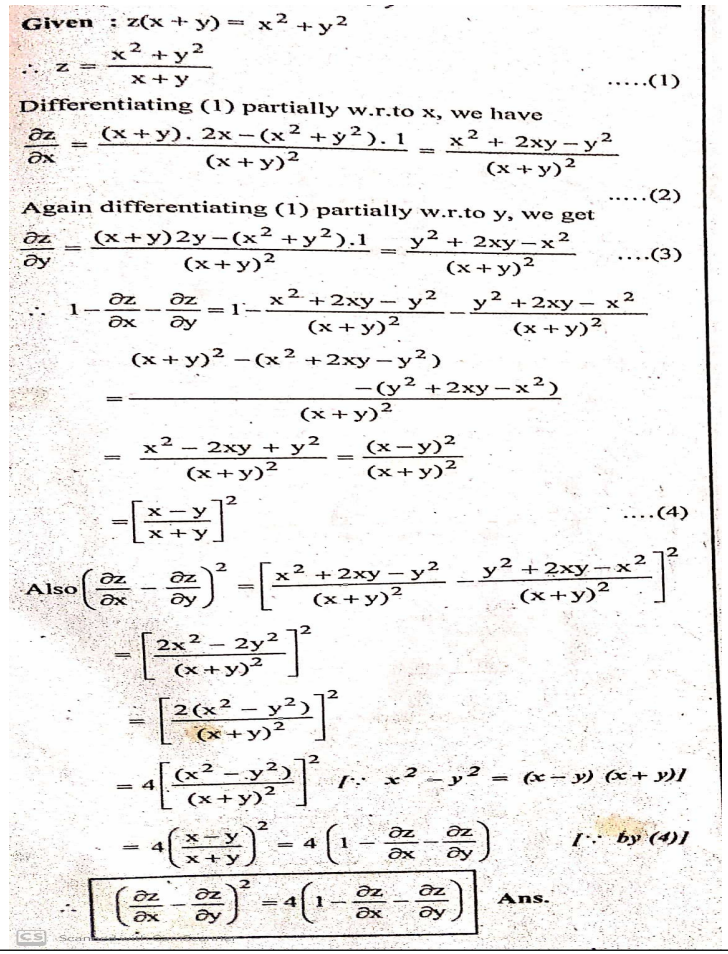
		$\Rightarrow x.\frac{\partial^2 u}{\partial x\,\partial y}+y\frac{\partial^2 u}{\partial y^2}=(2\cos 2u-1)\frac{\partial u}{\partial y}$ Multiplying with y, we get $xy\frac{\partial^2 u}{\partial x\,\partial y}+y^2\frac{\partial^2 u}{\partial y^2}=(2\cos 2u-1)y\frac{\partial u}{\partial y}$ .....(3) Adding (2) and (3), we get $x^2.\frac{\partial^2 u}{\partial x^2}+2xy\frac{\partial^2 u}{\partial x\,\partial y}+y^2\frac{\partial^2 u}{\partial y^2}$ $=(2\cos 2u-1)\left[x\frac{\partial u}{\partial x}+y\frac{\partial u}{\partial y}\right]$ $=(2\cos 2u-1)\sin 2u$ [∵ by (1)] $=2\sin 2u\cos 2u-\sin 2u$ $=\sin 4u-\sin 2u$ [∵ sin 4u = 2 sin 2u cos 2u] ∴ $\boxed{x^2\frac{\partial^2 u}{\partial x^2}+2xy\frac{\partial^2 u}{\partial x\,\partial y}+y^2\frac{\partial^2 u}{\partial y^2}=\sin 4u-\sin 2u}$ Ans. OR $x^2\frac{\partial^2 u}{\partial x^2}+2xy\frac{\partial^2 u}{\partial x\,\partial y}+y^2\frac{\partial^2 u}{\partial y^2}$ $=(2\cos 2u-1)\sin 2u$ $=[2(2\cos^2 u-1)-1]2\sin u\times\cos u$ $=2\sin u\,[4\cos^3 u-3\cos u]$ ∴ $\boxed{x^2\frac{\partial^2 u}{\partial x^2}+2xy\frac{\partial^2 u}{\partial x\,\partial y}+y^2\frac{\partial^2 u}{\partial y^2}=2\sin u\cos 3u}$ Ans. [∵ cos 3u = 4 cos³ u − 3 cos u]						
3	Unit-II	If, prove that . Sol: Given : $z=e^{ax+by}f(ax-by)$ Differentiating partially w. r. to x & y, we get $\frac{\partial z}{\partial x}=e^{ax+by}a\,f'(ax-by)+ae^{ax+by}f(ax-by)$ and $\frac{\partial z}{\partial y}=-be^{ax+by}f'(ax-by)+be^{ax+by}f(ax-by)$ ∴ $b\frac{\partial z}{\partial x}+a\frac{\partial z}{\partial y}=b[\,ae^{ax+by}(f'(ax-by)+f(ax-by))]$ $+a[be^{ax+by}(-f'(ax-by)+f(ax-by))]$ $=abe^{ax+by}[f'(ax-by)+f(ax-by)$ $-f'(ax-by)+f(ax-by)]$ $=abe^{ax+by}.2.f(ax-by)$ $=2ab[e^{ax+by}f(ax-by)]$ $=2abz$ [∵ $z=e^{ax+by}f(ax-by)$] ∴ $\boxed{b\frac{\partial z}{\partial x}+a\frac{\partial z}{\partial y}=2abz}$ Ans. which is the required result.	15	5	Low	Easy	CO-2	Descriptive
4	Unit-II	If u = f (x^2+2yz, y^2+2xz) then find the value of $(y^2-xz)\frac{\partial u}{\partial x}+(x^2-yz)\frac{\partial u}{\partial y}+(z^2-xy)\frac{\partial u}{\partial z}$	15	5	Low	Easy	CO-2	Descriptive

		Sol: **Given :** $u = f(x^2 + 2yz, y^2 + 2xz) = f(v, w)$ where, $v = x^2 + 2yz$ and $w = y^2 + 2xz$ $\therefore \; u \to v, \; w \to x, y$ $\Leftrightarrow \frac{\partial u}{\partial x} = \frac{\partial u}{\partial v}\frac{\partial v}{\partial x} + \frac{\partial u}{\partial w}\frac{\partial w}{\partial x} = 2x\frac{\partial u}{\partial v} + 2z\frac{\partial v}{\partial w}$ $\left[\because \frac{\partial v}{\partial x} = 2x \;\&\; \frac{\partial w}{\partial x} = 2z\right]$ and $\frac{\partial u}{\partial y} = \frac{\partial u}{\partial v}\frac{\partial v}{\partial y} + \frac{\partial u}{\partial w}\frac{\partial w}{\partial y} = 2z\frac{\partial u}{\partial v} + 2y\frac{\partial u}{\partial w}$ $\left[\because \frac{\partial v}{\partial y} = 2z \;\&\; \frac{\partial w}{\partial y} = 2y\right]$ Also $\frac{\partial u}{\partial z} = \frac{\partial u}{\partial v}\frac{\partial v}{\partial z} + \frac{\partial u}{\partial w}\frac{\partial w}{\partial z} = 2y\frac{\partial u}{\partial v} + 2x\frac{\partial v}{\partial w}$ $\left[\because \frac{\partial v}{\partial z} = 2y \;\&\; \frac{\partial w}{\partial z} = 2x\right]$ Consider, $(y^2 - xz)\frac{\partial u}{\partial x} + (x^2 - yz)\frac{\partial u}{\partial y} + (z^2 - xy)\frac{\partial u}{\partial z}$ $= (y^2 - xz)2\left(x\frac{\partial u}{\partial v} + z\frac{\partial u}{\partial w}\right)$ $+ (x^2 - yz)2\left(z\frac{\partial u}{\partial v} + y\frac{\partial v}{\partial w}\right)$ $+ (z^2 - xy)2\left(y\frac{\partial u}{\partial v} + x\frac{\partial v}{\partial w}\right)$ $= 2(xy^2 - zx^2 + x^2z - yz^2 + yz^2 - xy^2)\frac{\partial u}{\partial v}$ $+ 2(y^2z - z^2x + x^2y - y^2z + z^2x - x^2y)\frac{\partial u}{\partial w}$ $= 0$ $\therefore$ The required result is $\boxed{(y^2 - xz)\frac{\partial u}{\partial x} + (x^2 - yz)\frac{\partial u}{\partial y} + (z^2 - xy)\frac{\partial u}{\partial z} = 0}$ Ans.						
5	Unit-II	If $u(x, y) = \frac{x^2 + y^2}{x + y}$ then show that $\left(\frac{\partial u}{\partial x} - \frac{\partial u}{\partial y}\right)^2 = 4\left(1 - \frac{\partial u}{\partial x} - \frac{\partial u}{\partial y}\right)$	15	5	Medium	Moderate	CO-2	Descriptive

		Sol: 						
6	Unit-II	If, show that Sol:	15	5	Medium	Moderate	CO-2	Numerical

Given : $z(x+y) = x^2 + y^2$

$\therefore\ z = \dfrac{x^2+y^2}{x+y}$ .....(1)

Differentiating (1) partially w.r.to x, we have

$$\frac{\partial z}{\partial x} = \frac{(x+y).2x-(x^2+y^2).1}{(x+y)^2} = \frac{x^2+2xy-y^2}{(x+y)^2} \quad .....(2)$$

Again differentiating (1) partially w.r.to y, we get

$$\frac{\partial z}{\partial y} = \frac{(x+y)2y-(x^2+y^2).1}{(x+y)^2} = \frac{y^2+2xy-x^2}{(x+y)^2} \quad ....(3)$$

$$\therefore\ 1-\frac{\partial z}{\partial x}-\frac{\partial z}{\partial y} = 1-\frac{x^2+2xy-y^2}{(x+y)^2}-\frac{y^2+2xy-x^2}{(x+y)^2}$$

$$= \frac{(x+y)^2-(x^2+2xy-y^2)-(y^2+2xy-x^2)}{(x+y)^2}$$

$$= \frac{x^2-2xy+y^2}{(x+y)^2} = \frac{(x-y)^2}{(x+y)^2}$$

$$= \left[\frac{x-y}{x+y}\right]^2 \quad ....(4)$$

$$\text{Also } \left(\frac{\partial z}{\partial x}-\frac{\partial z}{\partial y}\right)^2 = \left[\frac{x^2+2xy-y^2}{(x+y)^2}-\frac{y^2+2xy-x^2}{(x+y)^2}\right]^2$$

$$= \left[\frac{2x^2-2y^2}{(x+y)^2}\right]^2$$

$$= \left[\frac{2(x^2-y^2)}{(x+y)^2}\right]^2$$

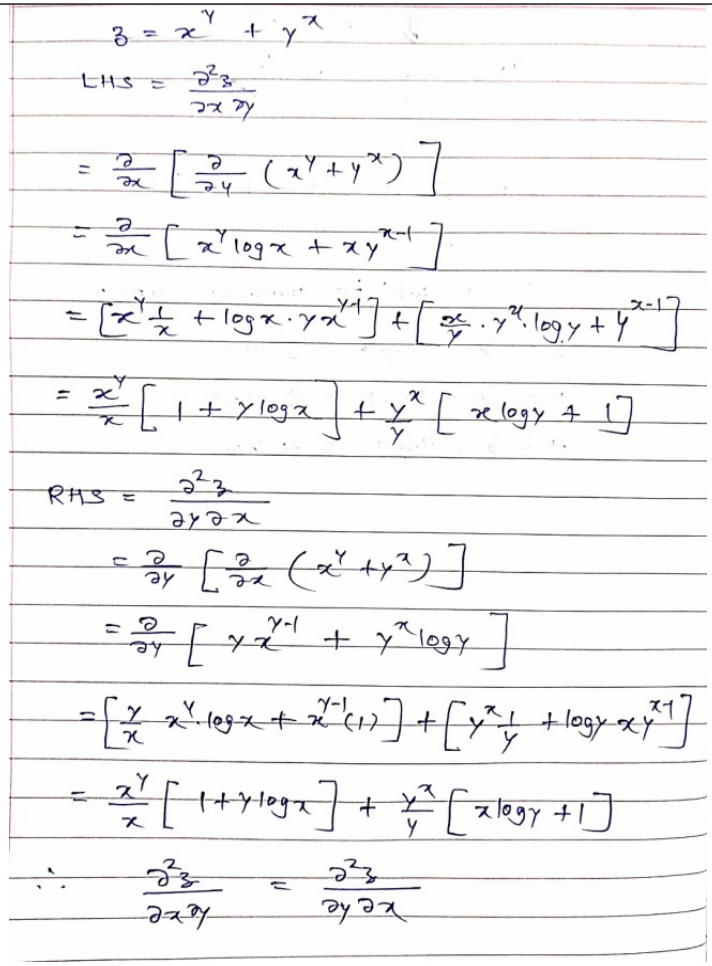
$$= 4\left[\frac{(x^2-y^2)}{(x+y)^2}\right]^2 \quad [\because\ x^2-y^2 = (x-y)(x+y)]$$

$$= 4\left(\frac{x-y}{x+y}\right)^2 = 4\left(1-\frac{\partial z}{\partial x}-\frac{\partial z}{\partial y}\right) \quad [\because \text{ by (4)}]$$

$$\therefore\ \boxed{\left(\frac{\partial z}{\partial x}-\frac{\partial z}{\partial y}\right)^2 = 4\left(1-\frac{\partial z}{\partial x}-\frac{\partial z}{\partial y}\right)} \quad \text{Ans.}$$

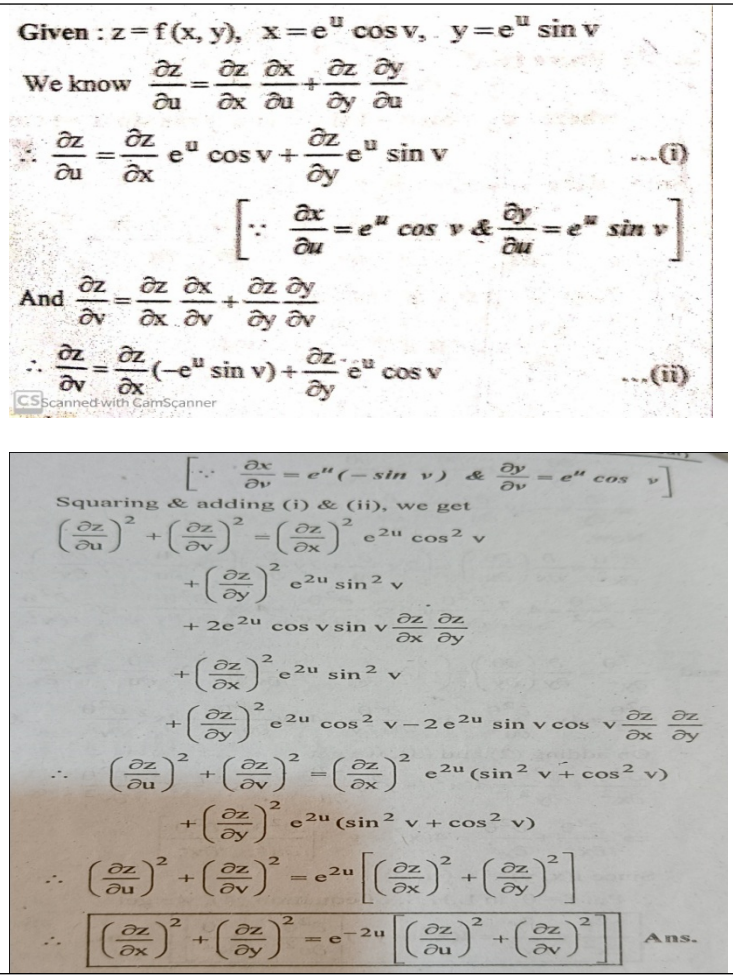
		Given : $u = \log(x^4 + x^2 y^2 + y^4)$ **Let** $z = e^u = (x^4 + x^2y^2 + y^4)$ $= x^4\left[1 + \left(\frac{y}{x}\right)^2 + \left(\frac{y}{x}\right)^4\right]$ $\Rightarrow \quad z = x^4 f\left(\frac{y}{x}\right)$ **Thus, z is a homogeneous function of degree 4.** ∴ **By Euler's theorem, we have** $x\frac{\partial z}{\partial x} + y\frac{\partial z}{\partial y} = nz$ **i.e.** $x\frac{\partial}{\partial x}(e^u) + y\frac{\partial}{\partial y}(e^u) = 4e^u$ $\therefore \quad xe^u\frac{\partial u}{\partial x} + ye^u\frac{\partial u}{\partial y} = 4e^u$ $\Rightarrow x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} = 4$ ....(1) **Differentiating equation (1) w.r. to x, we get** $x\frac{\partial^2 u}{\partial x^2} + \frac{\partial u}{\partial x} + y\frac{\partial^2 u}{\partial x\,\partial y} = 0$ **Multiplying by x, we get** $x^2\frac{\partial^2 u}{dx^2} + x\frac{\partial u}{\partial x} + xy\frac{\partial^2 u}{\partial x\,\partial y} = 0$ ...(2) **Again differentiating equation (1) w.r.to y, we get** $x\frac{\partial^2 u}{\partial x\,\partial y} + y\frac{\partial^2 u}{\partial y^2} + \frac{\partial u}{\partial y} = 0$ **Multiplying by y, we get** $xy\frac{\partial^2 u}{\partial x\,\partial y} + y^2\frac{\partial^2 u}{\partial y^2} + y\frac{\partial u}{\partial y} = 0$ ...(3) **Adding equation (2) and (3), we get** $x^2\frac{\partial^2 u}{\partial x^2} + x\frac{\partial u}{\partial x} + xy\frac{\partial^2 u}{\partial x\,\partial y} + xy\frac{\partial^2 u}{\partial x\,\partial y} + y^2\frac{\partial^2 u}{\partial y^2} + y\frac{\partial u}{\partial y} = 0$ $\Rightarrow x^2\frac{\partial^2 u}{\partial x^2} + 2xy\frac{\partial^2 u}{\partial x\,\partial y} + y^2\frac{\partial^2 u}{\partial y^2} + \left(x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y}\right) = 0$ ∴ **From equation (1), we get** $\boxed{x^2\frac{\partial^2 u}{\partial x^2} + 2xy\frac{\partial^2 u}{\partial x\,\partial y} + y^2\frac{\partial^2 u}{\partial y^2} + 4 = 0}$ **Ans.** Scanned with CamScanner						
7		If $u = f\left(\frac{y-x}{xy}, \frac{z-x}{zx}\right)$, Show that $x^2\frac{\partial u}{\partial x} + y^2\frac{\partial u}{\partial y} + z^2\frac{\partial u}{\partial z} = 0$	15	5	Medium	Moderate	CO-2	Descriptive

	Unit-II	Sol: Given : $u = f\left(\frac{y-x}{xy}, \frac{z-x}{xz}\right)$ Let $u = f(t_1, t_2)$ where $t_1 = \frac{y-x}{xy} = x^{-1} - y^{-1}$ and $t_2 = \frac{z-x}{xz} = x^{-1} - z^{-1}$ $\Rightarrow \frac{\partial t_1}{\partial x} = -\frac{1}{x^2}; \frac{\partial t_1}{\partial y} = \frac{1}{y^2}; \frac{\partial t_1}{\partial z} = 0$ and $\frac{\partial t_2}{\partial x} = -\frac{1}{x^2}; \frac{\partial t_2}{\partial y} = 0; \frac{\partial t_2}{\partial z} = \frac{1}{z^2}$ Now, by chain rule we have $\frac{\partial u}{\partial x} = \frac{\partial u}{\partial t_1}\frac{\partial t_1}{\partial x} + \frac{\partial u}{\partial t_2}\frac{\partial t_2}{\partial x}$ $= \frac{\partial u}{\partial t_1}\left(-\frac{1}{x^2}\right) + \frac{\partial u}{\partial t_2}\left(-\frac{1}{x^2}\right)$ $\Rightarrow x^2\frac{\partial u}{\partial x} = -\frac{\partial u}{\partial t_1} - \frac{\partial u}{\partial t_2}$ .....(1) And $\frac{\partial u}{\partial y} = \frac{\partial u}{\partial t_1}\frac{\partial t_1}{\partial y} + \frac{\partial u}{\partial t_2}\frac{\partial t_2}{\partial y} = \frac{\partial u}{\partial t_1}\left(\frac{1}{y^2}\right) + \frac{\partial u}{\partial t_2}(0)$ $\Rightarrow y^2\frac{\partial u}{\partial y} = \frac{\partial u}{\partial t_1}$ .....(2) Also, $\frac{\partial u}{\partial z} = \frac{\partial u}{\partial t_1}\frac{\partial t_1}{\partial z} + \frac{\partial u}{\partial t_2}\frac{\partial t_2}{\partial z} = \frac{\partial u}{\partial t_1}(0) + \frac{\partial u}{\partial t_2}\left(\frac{1}{z^2}\right)$ $\Rightarrow z^2\frac{\partial u}{\partial z} = \frac{\partial u}{\partial t_2}$ .....(3) Adding equation (1) and (2) and (3), we get Scanned with CamScanner $x^2\frac{\partial u}{\partial x} + y^2\frac{\partial u}{\partial y} + z^2\frac{\partial u}{\partial z} = -\frac{\partial u}{\partial t_1} - \frac{\partial u}{\partial t_2} + \frac{\partial u}{\partial t_1} + \frac{\partial u}{\partial t_2}$ $\Rightarrow x^2\frac{\partial u}{\partial x} + y^2\frac{\partial u}{\partial y} + z^2\frac{\partial u}{\partial z} = 0$ $\therefore$ The required result is $\boxed{x^2\frac{\partial u}{\partial x} + y^2\frac{\partial u}{\partial y} + z^2\frac{\partial u}{\partial z} = 0}$ Ans. Scanned with CamScanner						
8		If $z = x^y + y^x$, prove that $\frac{\partial^2 z}{\partial x \partial y} = \frac{\partial^2 z}{\partial y \partial x}$ Sol:	15	5	Medium	Difficult	CO-2	Descriptive

	Unit-II							
9	Unit-II	$x^x y^y z^z = c$ If , show that at x=y=z $\frac{\partial^2 z}{\partial x \partial y} = -(x \log ex)^{-1}$ Sol:	15	5	Medium	Difficult	CO-2	Descriptive

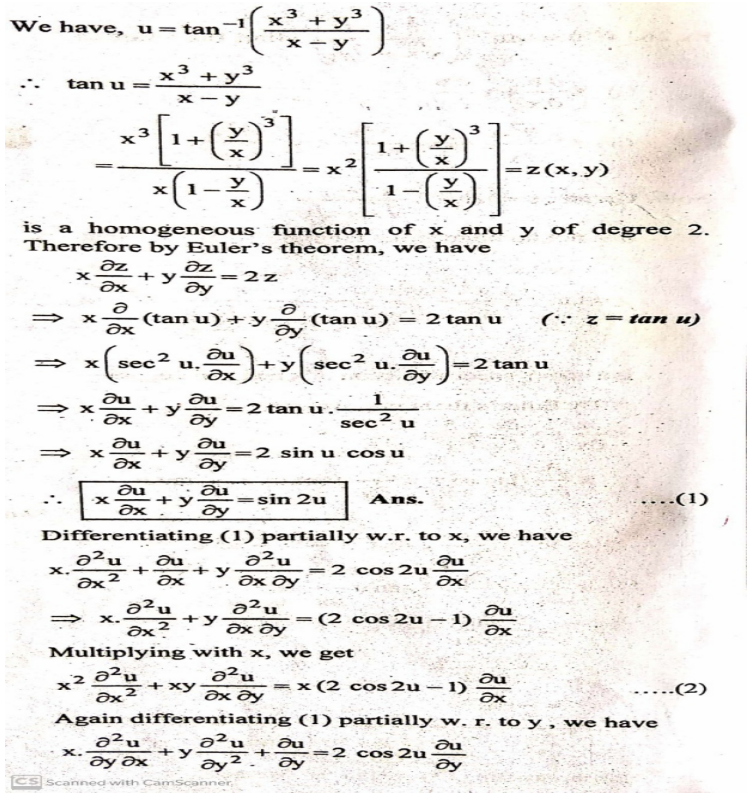
		Given : $x^x y^y z^z = c$ Taking log on both sides, we get $x \log x + y \log y + z \log z = \log c$ ....(i) Differentiating (i) partially w.r. to x and assuming z as a function of x and y, we have $\left(x.\frac{1}{x} + \log x\right) + 0 + \left(z.\frac{1}{z}\frac{\partial z}{\partial x} + \log z \frac{\partial z}{\partial x}\right) = 0$ $\Rightarrow \quad (1 + \log x) + (1 + \log z)\frac{\partial z}{\partial x} = 0$ $\Rightarrow \quad \frac{\partial z}{\partial x} = -\frac{1 + \log x}{1 + \log z}$ ....(ii) Similarly differentiating (i) partially w.r. to y , we have $\frac{\partial z}{\partial y} = -\frac{1 + \log y}{1 + \log z}$ .....(iii) Now differentiating (iii) partially w.r. to x , we get Scanned with CamScanner $\frac{\partial^2 z}{\partial x \partial y} = \frac{\partial}{\partial x}\left(\frac{\partial z}{\partial y}\right) = \frac{\partial}{\partial x}\left[-\frac{1 + \log y}{1 + \log z}\right]$ $= -(1 + \log y).\frac{-1}{(1 + \log z)^2}.\frac{1}{z}.\frac{\partial z}{\partial x}$ $= \frac{(1 + \log y)}{z(1 + \log z)^2}\left[-\frac{1 + \log x}{1 + \log z}\right]$ *[∵ from (ii)]* $\therefore \quad \frac{\partial^2 z}{\partial x \partial y} = -\frac{(1 + \log x)(1 + \log y)}{z(1 + \log z)^3}$ $\therefore$ at $x = y = z$, we have $\frac{\partial^2 z}{\partial x \partial y} = -\frac{(1 + \log x)(1 + \log x)}{x(1 + \log x)^3} = \frac{-1}{x(1 + \log x)}$ $= \frac{-1}{x(\log e + \log x)} = \frac{-1}{x \log ex}$ *[∵ log e = 1]* $\therefore \quad \boxed{\frac{\partial^2 z}{\partial x \partial y} = -(x \log ex)^{-1}}$ **Ans.** Scanned with CamScanner						
10	Unit-II	If, prove that i) ii) . Sol:	15	5	Medium	Difficult	CO-2	Descriptive

		Here $u = \log\left(\frac{x^4 - y^4}{x - y}\right)$ $\therefore e^u = \frac{x^4 - y^4}{x - y}$ is a homogeneous function of degree 3. $\therefore$ By Euler's theorem, we have $x\frac{\partial}{\partial x}e^u + y\frac{\partial}{\partial y}e^u = 3e^u$ $\Rightarrow xe^u\frac{\partial u}{\partial x} + ye^u\frac{\partial u}{\partial y} = 3e^u$ $\therefore \boxed{x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} = 3}$ **Ans.** ....(1) Differentiating (1) w.r. to x partially, we get $x\frac{\partial^2 u}{\partial x^2} + \frac{\partial u}{\partial x} + y\frac{\partial^2 u}{\partial x\,\partial y} = 0$ .....(2) Differentiating (1) partially w.r. to y, we get $x\frac{\partial^2 u}{\partial y\,\partial x} + y\frac{\partial^2 u}{\partial y^2} + \frac{\partial u}{\partial y} = 0$ ....(3) eq. (2) × x + eq.(3) × y $\Rightarrow$ $x^2\frac{\partial^2 u}{\partial x^2} + 2xy\frac{\partial^2 u}{\partial x\,\partial y} + y^2\frac{\partial^2 u}{\partial y^2} + x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} = 0$ $\therefore x^2\frac{\partial^2 u}{\partial x^2} + 2xy\frac{\partial^2 u}{\partial x\,\partial y} + y^2\frac{\partial^2 u}{\partial y^2} + 3 = 0$ *(∵ by (1))* $\therefore \boxed{x^2\frac{\partial^2 u}{\partial x^2} + 2xy\frac{\partial^2 u}{\partial x\,\partial y} + y^2\frac{\partial^2 u}{\partial y^2} = -3}$ **Ans.** which is the required result. Scanned with CamScanner						
11	Unit-II	If, where and, then show that Sol:	12	4	Medium	Difficult	CO-2	Descriptive

								
12	Unit-II	If u = log (tanx + tany + tan z) . Prove that $\sin 2x\frac{\partial u}{\partial x}+\sin 2y\frac{\partial u}{\partial y}+\sin 2z\frac{\partial u}{\partial z}=2$ Sol:	12	4	Medium	Difficult	CO-2	Descriptive

		Given : $u = \log(\tan x + \tan y + \tan z)$ $\Rightarrow \frac{\partial u}{\partial x} = \frac{\sec^2 x}{\tan x + \tan y + \tan z}$ $\frac{\partial u}{\partial y} = \frac{\sec^2 y}{\tan x + \tan y + \tan z}$ and $\frac{\partial u}{\partial z} = \frac{\sec^2 z}{\tan x + \tan y + \tan z}$ $\therefore \sin 2x \frac{\partial u}{\partial x} + \sin 2y \frac{\partial u}{\partial y} + \sin 2z \frac{\partial u}{\partial z}$ $= 2 \sin x \cos x \frac{\sec^2 x}{\tan x + \tan y + \tan z}$ $+ 2 \sin y \cos y \frac{\sec^2 y}{\tan x + \tan y + \tan z}$ $+ 2 \sin z \cos z \frac{\sec^2 z}{\tan x + \tan y + \tan z}$ $= 2 \tan x \frac{1}{\tan x + \tan y + \tan z}$ $+ 2 \tan y \frac{1}{\tan x + \tan y + \tan z}$ $+ 2 \tan z \frac{1}{\tan x + \tan y + \tan z}$ Scanned with CamScanner $= 2\left(\frac{\tan x + \tan y + \tan z}{\tan x + \tan y + \tan z}\right) = 2$ $\therefore \boxed{\sin 2x \frac{\partial u}{\partial x} + \sin 2y \frac{\partial u}{\partial y} + \sin 2z \frac{\partial u}{\partial z} = 2}$ Scanned with CamScanner						
13	Unit-II	If u= $\log\left(\frac{x^2+y^2}{\sqrt{x}+\sqrt{y}}\right)$ find the value of	12	4	Medium	Difficult	CO-2	Descriptive

		(i) (ii) Sol: **Given :** $u = \log\left(\frac{x^2+y^2}{\sqrt{x}+\sqrt{y}}\right)$ Let $z = e^u = \frac{x^2+y^2}{\sqrt{x}+\sqrt{y}} = x^{\frac{3}{2}} f\left(\frac{y}{x}\right)$ is a homogeneous function of degree $\frac{3}{2}$. ∴ By Euler's theorem, we have $x\frac{\partial z}{\partial x} + y\frac{\partial z}{\partial y} = \frac{3}{2}z$ $\Rightarrow x\frac{\partial}{\partial x}(e^u) + y\frac{\partial}{\partial y}(e^u) = \frac{3}{2}e^u$ $\Rightarrow x e^u \frac{\partial u}{\partial x} + y e^u \frac{\partial u}{\partial y} = \frac{3}{2}e^u$ $\therefore \boxed{x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} = \frac{3}{2}}$ **Ans.** .....(1) Differentiating (1) partially w. r. to x, we get $x\frac{\partial^2 u}{\partial x^2} + \frac{\partial u}{\partial x} + y\frac{\partial^2 u}{\partial x\,\partial y} = 0$ $\Rightarrow x^2\frac{\partial^2 u}{\partial x^2} + x\frac{\partial u}{\partial x} + xy\frac{\partial^2 u}{\partial x\,\partial y} = 0$ .....(2) Again differentiating (1) partially w.r. to y, we have $x\frac{\partial^2 u}{\partial y\,\partial x} + y\frac{\partial^2 u}{\partial y^2} + \frac{\partial u}{\partial y} = 0$ $\Rightarrow xy\frac{\partial^2 u}{\partial x\,\partial y} + y^2\frac{\partial^2 u}{\partial y^2} + y\frac{\partial u}{\partial y} = 0$ .....(3) Adding (2) and (3), we get $x^2\frac{\partial^2 u}{\partial x^2} + 2xy\frac{\partial^2 u}{\partial x\,\partial y} + y^2\frac{\partial^2 u}{\partial y^2} + \left(x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y}\right) = 0$ ∴ The required result is $\boxed{x^2\frac{\partial^2 u}{\partial x^2} + 2xy\frac{\partial^2 u}{\partial x\,\partial y} + y^2\frac{\partial^2 u}{\partial y^2} = -\frac{3}{2}}$ **Ans.** *(∵ by (1))* Scanned with CamScanner						
14	Unit-II	If $u = \tan^{-1}\left(\frac{x^3+y^3}{x-y}\right)$ prove that	12	4	Medium	Difficult	CO-2	Descriptive

		=sin 2u (1-4sin^{2}u) Sol: 						

We have, $u = \tan^{-1}\left(\frac{x^3+y^3}{x-y}\right)$

$\therefore \quad \tan u = \frac{x^3+y^3}{x-y}$

$$= \frac{x^3\left[1+\left(\frac{y}{x}\right)^3\right]}{x\left(1-\frac{y}{x}\right)} = x^2\left[\frac{1+\left(\frac{y}{x}\right)^3}{1-\left(\frac{y}{x}\right)}\right] = z(x,y)$$

is a homogeneous function of x and y of degree 2. Therefore by Euler's theorem, we have

$$x\frac{\partial z}{\partial x} + y\frac{\partial z}{\partial y} = 2z$$

$$\Rightarrow x\frac{\partial}{\partial x}(\tan u) + y\frac{\partial}{\partial y}(\tan u) = 2\tan u \qquad (\because z = \tan u)$$

$$\Rightarrow x\left(\sec^2 u.\frac{\partial u}{\partial x}\right) + y\left(\sec^2 u.\frac{\partial u}{\partial y}\right) = 2\tan u$$

$$\Rightarrow x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} = 2\tan u.\frac{1}{\sec^2 u}$$

$$\Rightarrow x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} = 2\sin u\cos u$$

$$\therefore \boxed{x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} = \sin 2u} \quad \text{Ans.} \qquad \ldots.(1)$$

Differentiating (1) partially w.r. to x, we have

$$x.\frac{\partial^2 u}{\partial x^2} + \frac{\partial u}{\partial x} + y\frac{\partial^2 u}{\partial x\,\partial y} = 2\cos 2u\frac{\partial u}{\partial x}$$

$$\Rightarrow x.\frac{\partial^2 u}{\partial x^2} + y\frac{\partial^2 u}{\partial x\,\partial y} = (2\cos 2u - 1)\frac{\partial u}{\partial x}$$

Multiplying with x, we get

$$x^2\frac{\partial^2 u}{\partial x^2} + xy\frac{\partial^2 u}{\partial x\,\partial y} = x(2\cos 2u - 1)\frac{\partial u}{\partial x} \qquad \ldots.(2)$$

Again differentiating (1) partially w. r. to y, we have

$$x.\frac{\partial^2 u}{\partial y\,\partial x} + y\frac{\partial^2 u}{\partial y^2} + \frac{\partial u}{\partial y} = 2\cos 2u\frac{\partial u}{\partial y}$$

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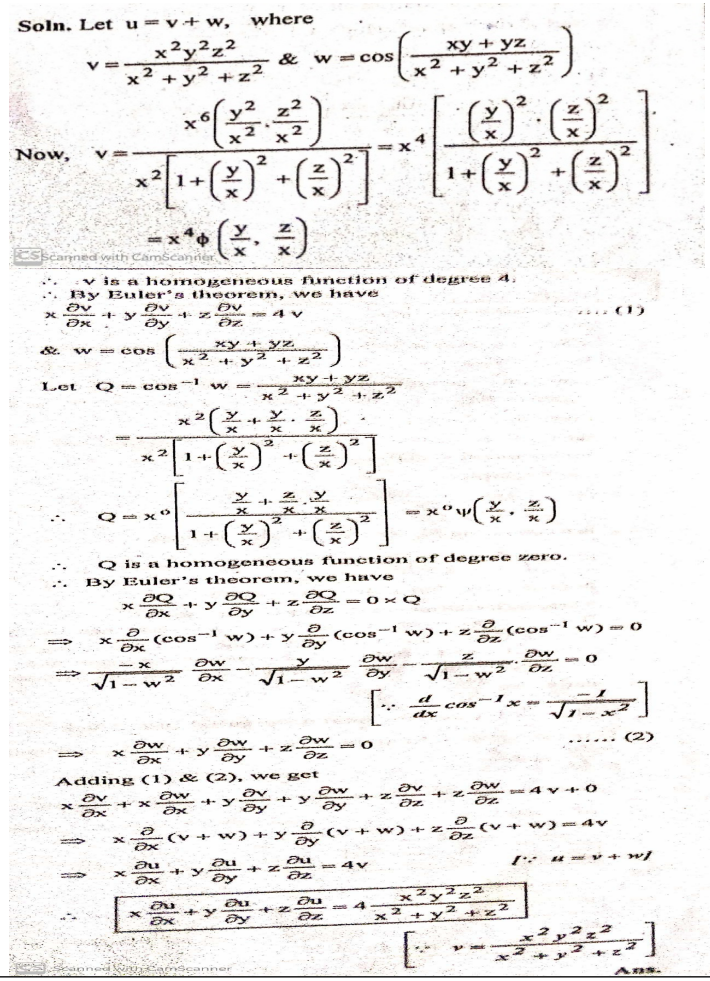
		$\Rightarrow x\cdot\frac{\partial^2 u}{\partial x\,\partial y}+y\frac{\partial^2 u}{\partial y^2}=(2\cos 2u-1)\frac{\partial u}{\partial y}$ Multiplying with y, we get $xy\frac{\partial^2 u}{\partial x\,\partial y}+y^2\frac{\partial^2 u}{\partial y^2}=(2\cos 2u-1)y\frac{\partial u}{\partial y}$ .....(3) Adding (2) and (3), we get $x^2\cdot\frac{\partial^2 u}{\partial x^2}+2xy\frac{\partial^2 u}{\partial x\,\partial y}+y^2\frac{\partial^2 u}{\partial y^2}$ $=(2\cos 2u-1)\left[x\frac{\partial u}{\partial x}+y\frac{\partial u}{\partial y}\right]$ $=(2\cos 2u-1)\sin 2u$ *[∵ by (1)]* $=2\sin 2u\cos 2u-\sin 2u$ $=\sin 4u-\sin 2u$ *[∵ sin 4u = 2 sin 2u cos 2u]* ∴ $\boxed{x^2\frac{\partial^2 u}{\partial x^2}+2xy\frac{\partial^2 u}{\partial x\,\partial y}+y^2\frac{\partial^2 u}{\partial y^2}=\sin 4u-\sin 2u}$ **Ans.** **OR** $x^2\frac{\partial^2 u}{\partial x^2}+2xy\frac{\partial^2 u}{\partial x\,\partial y}+y^2\frac{\partial^2 u}{\partial y^2}$ $=(2\cos 2u-1)\sin 2u$ $=[2(2\cos^2 u-1)-1]2\sin u\times\cos u$ $=2\sin u\,[4\cos^3 u-3\cos u]$ ∴ $\boxed{x^2\frac{\partial^2 u}{\partial x^2}+2xy\frac{\partial^2 u}{\partial x\,\partial y}+y^2\frac{\partial^2 u}{\partial y^2}=2\sin u\cos 3u}$ **Ans.** *[∵ cos 3u = 4 cos³ u − 3 cos u]* Scanned with CamScanner						
15	Unit-II	If, Show that	12	4	Medium	Difficult	CO-2	Descriptive

		Sol: 						
16	Unit-II	If, find the value of i) ii) . Sol:	15	4	high	Very difficult	CO-2	Descriptive

		(i) We are given $u = \tan^{-1}\left\{\frac{x^{\frac{1}{2}} + y^{\frac{1}{2}}}{x^{\frac{1}{3}} - y^{\frac{1}{3}}}\right\}$ $\therefore \tan u = \frac{x^{\frac{1}{2}} + y^{\frac{1}{2}}}{x^{\frac{1}{3}} - y^{\frac{1}{3}}}$ $= \frac{x^{1/2}[1 + (y/x)^{1/2}]}{x^{1/3}[1 - (y/x)^{1/3}]} = x^{\frac{1}{2} - \frac{1}{3}}\,\phi\left(\frac{y}{x}\right)$ is a homogeneous function of degree $\frac{1}{2} - \frac{1}{3} = \frac{1}{6}$. Hence by Euler's theorem, we have $x\frac{\partial}{\partial x}(\tan u) + y\frac{\partial}{\partial y}(\tan u) = \frac{1}{6}\tan u$ $\Rightarrow x\sec^2 u\frac{\partial u}{\partial x} + y\sec^2 u\frac{\partial u}{\partial y} = \frac{1}{6}\tan u$ $\Rightarrow x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} = \frac{1}{6}\frac{\sin u}{\cos u}\cos^2 u$ $= \frac{1}{6}\sin u\cos u$ $= \frac{1}{12}\sin 2u$ $\Rightarrow x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} = \frac{1}{12}.\sin 2u$ .....(1) $\therefore$ The required result is $\boxed{x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} = \frac{1}{12}.\sin 2u}$ Ans. (ii) Differentiating (1) partially w.r. to x , we get $x\frac{\partial^2 u}{\partial x^2} + \frac{\partial u}{\partial x} + y\frac{\partial^2 u}{\partial x\partial y} = \frac{2}{12}\cos 2u\frac{\partial u}{\partial x}$ ....(2) Differentiating (1) partially w.r.to y, we get $x\frac{\partial^2 u}{\partial x\partial y} + y\frac{\partial^2 u}{\partial y^2} + \frac{\partial u}{\partial y} = \frac{2}{12}\cos 2u\frac{\partial u}{\partial y}$ ....(3) Now, (2) x + (3) y $\Rightarrow$ $x^2\frac{\partial^2 u}{\partial x^2} + 2xy\frac{\partial^2 u}{\partial x\partial y} + y^2\frac{\partial^2 u}{\partial y^2}$ Scanned with CamScanner $= \left(\frac{2}{12}\cos 2u - 1\right)\left(x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y}\right)$ $= \frac{1}{12}\sin 2u\left(\frac{1}{6}\cos 2u - 1\right)$ (∵ by (1)) $\therefore$ The required results is $\boxed{x^2\frac{\partial^2 u}{\partial x^2} + 2xy\frac{\partial^2 u}{\partial x\,\partial y} + y^2\frac{\partial^2 u}{\partial y^2} = \frac{1}{12}\sin 2u\left(\frac{1}{6}\cos 2u - 1\right)}$ Scanned with CamScanner						
17	Unit-II	If, prove that: Sol: **Soln.** Let $z = \text{cosec } u = \left[\frac{x^{1/2} + y^{1/2}}{x^{1/3} + y^{1/3}}\right]^{1/2}$ $= \left[\frac{x^{1/2}\left(1 + \left(\frac{y}{x}\right)^{1/2}\right)}{x^{1/3}\left(1 + \left(\frac{y}{x}\right)^{1/3}\right)}\right]^{1/2}$ Scanned with CamScanner	15	4	high	Very difficult	CO-2	Descriptive

$$= \left[x^{\frac{1}{2}-\frac{1}{3}}\right]^{\frac{1}{2}} \left[\frac{1+\left(\frac{y}{x}\right)^{1/2}}{1+\left(\frac{y}{x}\right)^{1/3}}\right]^{1/2}$$

$$= x^{1/12}\phi\left(\frac{y}{x}\right)$$

$\therefore$ z is a homogeneous function of degree $\frac{1}{12}$

$\therefore$ By Euler's theorem, we have

$$x\frac{\partial z}{\partial x} + y\frac{\partial x}{\partial y} = \frac{1}{12}z$$

$$\Rightarrow x\frac{\partial}{\partial x}(\text{cosec } u) + y\frac{\partial}{\partial y}(\text{cosec } u) = \frac{1}{12}(\text{cosec } u)$$

$$\Rightarrow -\text{cosec } u \cot u\, x\frac{\partial u}{\partial x} - \text{cosec } u \cot u\, y\frac{\partial u}{\partial y}$$

$$= \frac{1}{12}\text{cosec } u$$

$$\Rightarrow x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} = -\frac{1}{12}\frac{1}{\cot u}$$

$$\Rightarrow x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} = -\frac{1}{12}\tan u \quad \ldots\ldots (1)$$

Differentiating (1) w. r. to x, we get

$$x\frac{\partial^2 u}{\partial x^2} + y\frac{\partial^2 u}{\partial x\,\partial y} + \frac{\partial u}{\partial x} = -\frac{1}{12}\sec^2 u\frac{\partial u}{\partial x}$$

Multiplying above equation by x, we get

$$x^2\frac{\partial^2 u}{\partial x^2} + xy\frac{\partial^2 u}{\partial x\,\partial y} + x\frac{\partial u}{\partial x} = -\frac{1}{12}\sec^2 u \quad x\frac{\partial u}{\partial x} \quad \ldots\ldots (2)$$

Now differentiate (1) w. r. to y, we get

$$x\frac{\partial^2 u}{\partial x\,\partial y} + y\frac{\partial^2 u}{\partial y^2} + \frac{\partial u}{\partial y} = -\frac{1}{12}\sec^2 u\frac{\partial u}{\partial y}$$

Multiplying above equation by y, we get

$$xy\frac{\partial^2 u}{\partial x\,\partial y} + y^2\frac{\partial^2 u}{\partial y^2} + y\frac{\partial u}{\partial y} = -\frac{1}{12}\sec^2 u \quad y\frac{\partial u}{\partial y} \quad \ldots\ldots (3)$$

Adding (2) & (3), we have

$$x^2\frac{\partial^2 u}{\partial x^2} + 2xy\frac{\partial^2 u}{\partial x\,\partial y} + y^2\frac{\partial^2 u}{\partial y^2} + \left(x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y}\right)$$

$$= -\frac{1}{12}\sec^2 u\left(x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y}\right)$$

$$\Rightarrow x^2\frac{\partial^2 u}{\partial x^2} + 2xy\frac{\partial^2 u}{\partial x\,\partial y} + y^2\frac{\partial^2 u}{\partial y^2}$$

$$= \left(-\frac{1}{12}\sec^2 u - 1\right)\left(x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y}\right)$$

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		$= -\left(1+\frac{1}{12}\sec^2 u\right)\left(-\frac{1}{12}\tan u\right)$ (∵ by (1)) $= \frac{1}{12}\tan u\left[1+\frac{1}{12}(1+\tan^2 u)\right]$ [∵ $\sec^2 u = 1+\tan^2 u$] $= \frac{1}{12}\tan u\left(\frac{13}{12}+\frac{1}{12}\tan^2 u\right)$ $= \frac{1}{144}\tan u[13+\tan^2 u]$ ∴ The required result is $x^2\frac{\partial^2 u}{\partial x^2}+2xy\frac{\partial^2 u}{\partial x\,\partial y}+y^2\frac{\partial^2 u}{\partial y^2}=\frac{1}{144}\tan u[13+\tan^2 u]$						
18	Unit-II	If, where and, then show that	15	4	high	Very difficult	CO-2	Descriptive

Sol:

Here $u = f(x, y)$ & $x = e^r \cos\theta$, $y = e^r \sin\theta$

and $u \to x, y \to r, \theta$,

$$\Rightarrow \frac{\partial u}{\partial r} = \frac{\partial u}{\partial x}\frac{\partial x}{\partial r} + \frac{\partial u}{\partial y}\frac{\partial y}{\partial r} = \frac{\partial u}{\partial x} e^r \cos\theta + \frac{\partial u}{\partial y} e^r \sin\theta$$

$$\left[\because \frac{\partial x}{\partial r} = e^r \cos\theta, \quad \frac{\partial y}{\partial r} = e^r \sin\theta\right]$$

$$\therefore \quad \frac{\partial}{\partial r} = \frac{\partial}{\partial x} e^r \cos\theta + \frac{\partial}{\partial y} e^r \sin\theta \qquad \ldots(1)$$

$$\Rightarrow \frac{\partial^2 u}{\partial r^2} = \frac{\partial}{\partial r}\left(\frac{\partial u}{\partial r}\right)$$

$$= \left(\frac{\partial}{\partial x} e^r \cos\theta + \frac{\partial}{\partial y} e^r \sin\theta\right)$$

$$\times \left(\frac{\partial u}{\partial x} e^r \cos\theta + \frac{\partial u}{\partial y} e^r \sin\theta\right) \qquad [\because \textit{from (1)}]$$

$$\Rightarrow \frac{\partial^2 u}{\partial r^2} = e^{2r} \cos^2\theta \frac{\partial^2 u}{\partial x^2} + e^{2r} \sin\theta\cos\theta \frac{\partial^2 u}{\partial x\,\partial y}$$

$$+ e^{2r} \sin\theta\cos\theta \frac{\partial^2 u}{\partial x\,\partial y} + e^{2r} \sin^2\theta \frac{\partial^2 u}{\partial y^2} \qquad \ldots(2)$$

and $$\frac{\partial u}{\partial \theta} = \frac{\partial u}{\partial x}\frac{\partial x}{\partial \theta} + \frac{\partial u}{\partial y}\frac{\partial y}{\partial \theta}$$

$$= \frac{\partial u}{\partial x}(-e^r \sin\theta) + \frac{\partial u}{\partial y} e^r \cos\theta$$

$$\left[\because \frac{\partial x}{\partial \theta} = -e^r \sin\theta, \quad \frac{\partial y}{\partial \theta} = e^r \cos\theta\right]$$

$$\therefore \quad \frac{\partial}{\partial \theta} = \frac{\partial}{\partial x} e^r \sin\theta + \frac{\partial}{\partial y} e^r \cos\theta \qquad \ldots(3)$$

$$\Rightarrow \frac{\partial^2 u}{\partial \theta^2} = \frac{\partial}{\partial \theta}\left(\frac{\partial u}{\partial \theta}\right) = \left(-\frac{\partial}{\partial x} e^r \sin\theta + \frac{\partial}{\partial y} e^r \cos\theta\right)$$

$$\times \left(-\frac{\partial u}{\partial x} e^r \sin\theta + \frac{\partial u}{\partial y} e^r \cos\theta\right) \qquad [\because \textit{from (2)}]$$

$$\Rightarrow \frac{\partial^2 u}{\partial \theta^2} = e^{2r} \sin^2\theta \frac{\partial^2 u}{\partial x^2} - e^{2r} \sin\theta\cos\theta \frac{\partial^2 u}{\partial x\,\partial y}$$

$$- e^{2r} \sin\theta \cos\theta \frac{\partial^2 u}{\partial y\,\partial x} + e^{2r} \cos^2\theta \frac{\partial^2 u}{\partial y^2} \qquad \ldots(4)$$

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$$\therefore \quad \frac{\partial^2 u}{\partial r^2}+\frac{\partial^2 u}{\partial \theta^2}=e^{2r}\frac{\partial^2 u}{\partial x^2}(\sin^2\theta+\cos^2\theta)$$

[∵ from (2) & (4)]

$$+e^{2r}\frac{\partial^2 u}{\partial y^2}(\sin^2\theta+\cos^2\theta)$$

$$=e^{2r}\left(\frac{\partial^2 u}{\partial x^2}+\frac{\partial^2 u}{\partial y^2}\right)$$

$$\therefore \quad \frac{\partial^2 u}{\partial x^2}+\frac{\partial^2 u}{\partial y^2}=\frac{1}{e^{2r}}\left(\frac{\partial^2 u}{\partial r^2}+\frac{\partial^2 u}{\partial \theta^2}\right)$$

$$\therefore \quad \boxed{\frac{\partial^2 u}{\partial x^2}+\frac{\partial^2 u}{\partial y^2}=e^{-2r}\left(\frac{\partial^2 u}{\partial r^2}+\frac{\partial^2 u}{\partial \theta^2}\right)} \quad \textbf{Ans.}$$

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19	Unit-II	If and, then show that	15	4	high	Very difficult	CO-2	Descriptive	

Sol:

Given: $x+y=2e^{\theta}\cos\phi$ and $x-y=2i\,e^{\theta}\sin\phi$ &
assume $u=f(x,y)$
$\therefore \quad u\to x, y\to \theta,\phi$

We have $\frac{\partial u}{\partial \theta}=\frac{\partial u}{\partial x}\cdot\frac{\partial x}{\partial \theta}+\frac{\partial u}{\partial y}\cdot\frac{\partial y}{\partial \theta}$ ...(1)

and $\frac{\partial u}{\partial \phi}=\frac{\partial u}{\partial x}\cdot\frac{\partial x}{\partial \phi}+\frac{\partial u}{\partial y}\cdot\frac{\partial y}{\partial \phi}$ ...(2)

From the given relations on adding and subtracting, we have-

$2x=2e^{\theta}(\cos\phi+i\sin\phi)$

$\Rightarrow \quad x=e^{\theta}.e^{i\phi}=e^{\theta+i\phi}$

$\therefore \quad \frac{\partial x}{\partial \theta}=e^{\theta+i\phi}=x$ and $\frac{\partial x}{\partial \phi}=i\,e^{\theta+i\phi}=ix$

and $2y=2e^{\theta}(\cos\phi-i\sin\phi)$

$\Rightarrow \quad y=e^{\theta}.e^{-i\phi}=e^{\theta-i\phi}$

$\therefore \quad \frac{\partial y}{\partial \theta}=e^{\theta-i\phi}=y$ and $\frac{\partial y}{\partial \phi}=-i\,e^{\theta-i\phi}=-iy$

then (1) $\Rightarrow \quad \frac{\partial u}{\partial \theta}=x\frac{\partial u}{\partial x}+y\frac{\partial u}{\partial y}$

$\therefore \quad \frac{\partial}{\partial \theta}=x\frac{\partial}{\partial x}+y\frac{\partial}{\partial y}$

Now $\frac{\partial^2 u}{\partial \theta^2}=\frac{\partial}{\partial \theta}\left(\frac{\partial u}{\partial \theta}\right)=\left(x\frac{\partial}{\partial x}+y\frac{\partial}{\partial y}\right)\left(x\frac{\partial u}{\partial x}+y\frac{\partial u}{\partial y}\right)$

$\Rightarrow \frac{\partial^2 u}{\partial \theta^2}=x\frac{\partial}{\partial x}\left(x\frac{\partial u}{\partial x}+y\frac{\partial u}{\partial y}\right)+y\frac{\partial}{\partial y}\left(x\frac{\partial u}{\partial x}+y\frac{\partial u}{\partial y}\right)$

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$$= x\left(x\frac{\partial^2 u}{\partial x^2} + \frac{\partial u}{\partial x} + y\frac{\partial^2 u}{\partial x\,\partial y}\right) + y\left(x\frac{\partial^2 u}{\partial y\,\partial x} + y\frac{\partial^2 u}{\partial y^2} + \frac{\partial u}{\partial y}\right)$$

$$\therefore \frac{\partial^2 u}{\partial \theta^2} = x^2\frac{\partial^2 u}{\partial x^2} + x\frac{\partial u}{\partial x} + 2xy\frac{\partial^2 u}{\partial x\,\partial y} + y^2\frac{\partial^2 u}{\partial y^2} + y\frac{\partial u}{\partial y} \quad ...(3)$$

$$(2) \Rightarrow \frac{\partial u}{\partial \phi} = ix\frac{\partial u}{\partial x} - iy\frac{\partial u}{\partial y}$$

$$\therefore \frac{\partial}{\partial \phi} = ix\frac{\partial}{\partial x} - iy\frac{\partial}{\partial y}$$

Now, $\frac{\partial^2 u}{\partial \phi^2} = \frac{\partial}{\partial \phi}\left(\frac{\partial u}{\partial \phi}\right)$

$$= \left(ix\frac{\partial}{\partial x} - iy\frac{\partial}{\partial y}\right)\left(ix\frac{\partial u}{\partial x} - iy\frac{\partial u}{\partial y}\right)$$

$$= ix\frac{\partial}{\partial x}\left(ix\frac{\partial u}{\partial x} - iy\frac{\partial u}{\partial y}\right) - iy\frac{\partial}{\partial y}\left(ix\frac{\partial u}{\partial x} - iy\frac{\partial u}{\partial y}\right)$$

$$= -x\left[x\frac{\partial^2 u}{\partial x^2} + \frac{\partial u}{\partial x}\right] + 2xy\frac{\partial^2 u}{\partial x\,\partial y} - y\left[y\frac{\partial^2 u}{\partial y^2} + \frac{\partial u}{\partial y}\right]$$

$$\therefore \frac{\partial^2 u}{\partial \phi^2} = -x^2\frac{\partial^2 u}{\partial x^2} - x\frac{\partial u}{\partial x} + 2xy\frac{\partial^2 u}{\partial x\,\partial y} - y^2\frac{\partial^2 u}{\partial y^2} - y\frac{\partial u}{\partial y} \quad ....(4)$$

Adding (3) and (4), we get

$$\boxed{\frac{\partial^2 u}{\partial \theta^2} + \frac{\partial^2 u}{\partial \phi^2} = 4xy\frac{\partial^2 u}{\partial x\,\partial y}}$$ **Ans.**

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20	Unit-II	If and, then show that Sol: We have $\phi = f(x, y, z)$ where x, y, & z are functions of u, v & w. Since $x = \sqrt{vw} \Rightarrow \frac{\partial x}{\partial u} = 0, \frac{\partial x}{\partial v} = \frac{w}{2\sqrt{vw}}, \frac{\partial x}{\partial w} = \frac{v}{2\sqrt{vw}}$ $y = \sqrt{uw} \Rightarrow \frac{\partial y}{\partial u} = \frac{w}{2\sqrt{uw}}, \frac{\partial y}{\partial v} = 0, \frac{\partial y}{\partial w} = \frac{u}{2\sqrt{uw}}$ $z = \sqrt{uv} \Rightarrow \frac{\partial z}{\partial u} = \frac{v}{2\sqrt{uv}}, \frac{\partial z}{\partial v} = \frac{u}{2\sqrt{uv}}, \frac{\partial z}{\partial w} = 0$ Then by using chain rule, we have $\frac{\partial \phi}{\partial u} = \frac{\partial \phi}{\partial x}\frac{\partial x}{\partial u} + \frac{\partial \phi}{\partial y}\frac{\partial y}{\partial u} + \frac{\partial \phi}{\partial z}\frac{\partial z}{\partial u}$ $\Rightarrow \frac{\partial \phi}{\partial u} = \frac{w}{2\sqrt{uw}}\frac{\partial \phi}{\partial y} + \frac{v}{2\sqrt{uv}}\frac{\partial \phi}{\partial z}$ ..... (1) $\frac{\partial \phi}{\partial v} = \frac{\partial \phi}{\partial x}\frac{\partial x}{\partial v} + \frac{\partial \phi}{\partial y}\frac{\partial y}{\partial v} + \frac{\partial \phi}{\partial z}\frac{\partial z}{\partial v}$ Scanned with CamScanner	15	4	high	Very difficult	CO-2	Descriptive

$$= x\left(x\frac{\partial^2 u}{\partial x^2}+\frac{\partial u}{\partial x}+y\frac{\partial^2 u}{\partial x\,\partial y}\right)+y\left(x\frac{\partial^2 u}{\partial y\,\partial x}+y\frac{\partial^2 u}{\partial y^2}+\frac{\partial u}{\partial y}\right)$$

$$\therefore \frac{\partial^2 u}{\partial \theta^2}=x^2\frac{\partial^2 u}{\partial x^2}+x\frac{\partial u}{\partial x}+2xy\frac{\partial^2 u}{\partial x\,\partial y}+y^2\frac{\partial^2 u}{\partial y^2}+y\frac{\partial u}{\partial y} \quad ...(3)$$

$$(2) \Rightarrow \frac{\partial u}{\partial \phi}=ix\frac{\partial u}{\partial x}-iy\frac{\partial u}{\partial y}$$

$$\therefore \frac{\partial}{\partial \phi}=ix\frac{\partial}{\partial x}-iy\frac{\partial}{\partial y}$$

Now, $$\frac{\partial^2 u}{\partial \phi^2}=\frac{\partial}{\partial \phi}\left(\frac{\partial u}{\partial \phi}\right)$$

$$=\left(ix\frac{\partial}{\partial x}-iy\frac{\partial}{\partial y}\right)\left(ix\frac{\partial u}{\partial x}-iy\frac{\partial u}{\partial y}\right)$$

$$=ix\frac{\partial}{\partial x}\left(ix\frac{\partial u}{\partial x}-iy\frac{\partial u}{\partial y}\right)-iy\frac{\partial}{\partial y}\left(ix\frac{\partial u}{\partial x}-iy\frac{\partial u}{\partial y}\right)$$

$$=-x\left[x\frac{\partial^2 u}{\partial x^2}+\frac{\partial u}{\partial x}\right]+2xy\frac{\partial^2 u}{\partial x\,\partial y}-y\left[y\frac{\partial^2 u}{\partial y^2}+\frac{\partial u}{\partial y}\right]$$

$$\therefore \frac{\partial^2 u}{\partial \phi^2}=-x^2\frac{\partial^2 u}{\partial x^2}-x\frac{\partial u}{\partial x}+2xy\frac{\partial^2 u}{\partial x\,\partial y}-y^2\frac{\partial^2 u}{\partial y^2}-y\frac{\partial u}{\partial y} \quad ....(4)$$

Adding (3) and (4), we get

$$\boxed{\frac{\partial^2 u}{\partial \theta^2}+\frac{\partial^2 u}{\partial \phi^2}=4xy\frac{\partial^2 u}{\partial x\,\partial y}}$$ **Ans.**

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